CIE AS PHYSICS (9702) • QUESTION PAPER

9.1 Electric Current

Syllabus Section: 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 20 Questions (20 Marks)

Question 1 Easy [1 mark]

When there is **no** current in a wire, which statement about the conduction electrons in that wire is correct?

A electrons in the wire are moving totally randomly within the wire

B equal numbers of electrons move at the same speed, but in opposite directions, along the wire

C no current is flowing therefore the electrons in the wire are stationary

D no current is flowing therefore the electrons in the wire are vibrating around a fixed point

Question 2 Easy [1 mark]

When will 1 C of charge pass a point in an electrical circuit?

A when 1 A moves through a potential difference of 1 V

B when a power of 1 W is used for 1 s

C when the current is 5 mA for 200 s

D when the current is 10 A for 10 s

Question 3 Easy [1 mark]

There is a current of 10 mA in a conductor for half an hour.

How much charge passes a point in the conductor in this time?

A 0.3 C

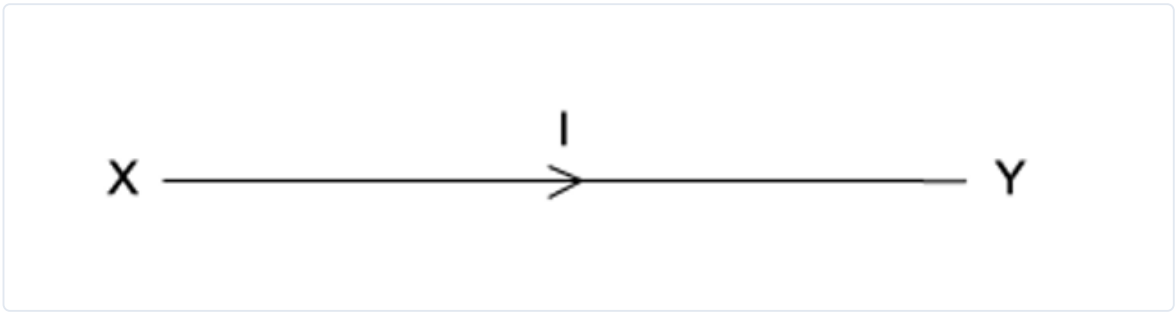
B 5 C

C 18 C

D 300 C

Question 4 Easy [1 mark]

The diagram shows the symbol for a wire carrying a current *I*



What does this current represent?

A the amount of charge flowing past a point in XY per second

B the number of electrons flowing past a point in XY per second

C the number of positive ions flowing past a point in XY per second

D the number of protons flowing past a point in XY per second

Question 5 Easy [1 mark]

Which equation that links some of the following terms is correct?

potential difference (p.d.)	*V*
current	*I*
resistance	*R*
charge	*Q*
energy	*E*
power	*P*
time	*t*

A $P = \frac{Q^2 R}{t}$

B $ER^2 = V^2 t$

C $\frac{VI}{P} = t$

D $PQ = EI$

Question 6 Easy [1 mark]

A copper wire of cross-sectional area 2.0 mm^2 carries a current of 10 A.

How many electrons pass through a given cross-section of the wire in one second?

A 1.0×10^1

B 5.0×10^6

C 6.3×10^{19}

D 3.1×10^{25}

Question 7 Medium [1 mark]

Two copper wires of the same length but different diameters carry the same current.

Which statement about the flow of charged particles through the wires is correct?

A charged particles are provided by the power supply, therefore, the speed at which they travel depends only on the voltage of the supply

B the charged particles in both wires move with the same average speed because the current in both wires is the same

C the charged particles move faster through the wire with the larger diameter because there is a greater volume through which to flow

D the charged particles move faster through the wire with the smaller diameter because it has a larger potential difference applied to it

Question 8 Medium [1 mark]

A milliammeter shows a reading of 20 mA.

How many electrons flow through the milliammeter in 10 seconds?

A 0.20

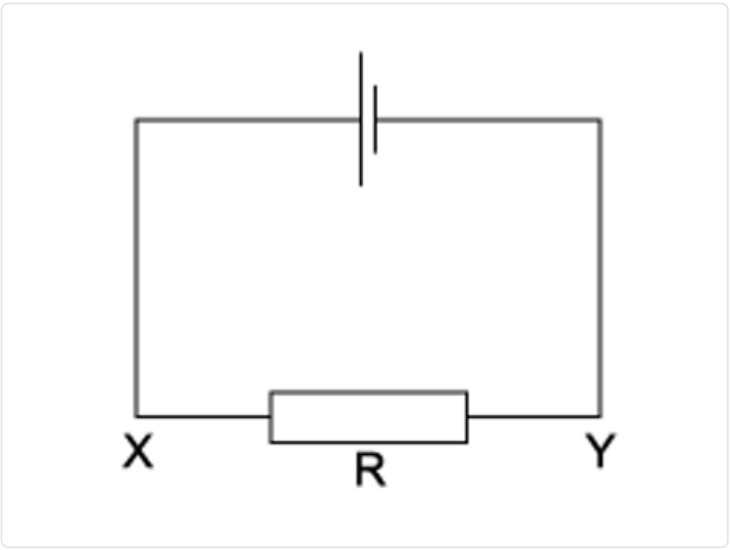
B 3.2×10^{20}

C 200

D 1.3×10^{18}

Question 9 Medium [1 mark]

The current in the circuit shown is 4.8 A.



What is the direction of flow and the rate of flow of electrons through the resistor *R* ?

	Direction of flow	Rate of flow
A	X to Y	$3.0 \times 10^{19}\ s^{-1}$
B	X to Y	$6.0 \times 10^{18}\ s^{-1}$
C	Y to X	$3.0 \times 10^{19}\ s^{-1}$
D	Y to X	$6.0 \times 10^{18}\ s^{-1}$

A

B

C

D

Question 10 Medium [1 mark]

A wire with a resistance of 2.0 Ω has 8.2×10^{28} electrons pass through it at an average drift speed of 0.09 mm s^{-1}. The potential difference of the resistor is 12 V.

What is the cross-sectional area of the wire?

A $5.1\ \mu m^2$

B $0.14\ mm^2$

C $5.1\ mm^2$

D $0.14\ cm^2$

Question 11 Medium [1 mark]

A 3.0 Ω resistor has 7.5×10^{18} electrons pass through it in 5.0 s.

What is the potential difference and charge of the resistor?

	Charge / C	Potential difference / V
A.	1.2	0.24
B.	1.2	0.72
C.	12	2.4
D.	12	7.2

A

B

C

D

Question 12 Medium [1 mark]

A beam of electrons, each carrying charge *e*, moves through a vacuum tube of uniform cross-sectional area *A* with uniform velocity *v*. The beam produces a current *I*.

What is the number density of electrons in the beam?

A $\frac{I}{Ae}$

B $\frac{I}{Ave}$

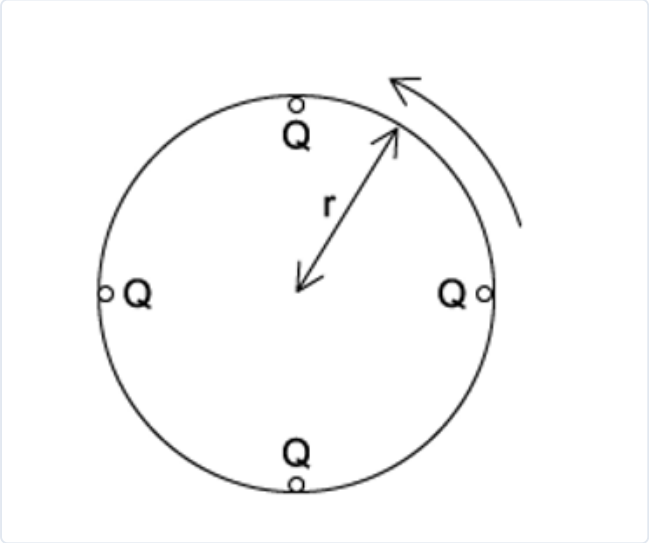
C $\frac{Iv}{Ae}$

D $\frac{I}{Av}$

Question 13 Hard [1 mark]

Four point charges, each of charge *Q*, are placed on the edge of an insulating disc of radius *r*.

The frequency of rotation of the disc is *f*.



What is the equivalent electric current at the edge of the disc ?

A $4Qf$

B $\frac{4Q}{f}$

C $8\pi rQ \quad f$

D $\frac{2Qf}{\pi r}$

Question 14 Hard [1 mark]

A parallel beam of protons, each with charge *e*, travel at uniform velocity *v*. This results in current *I* being formed.

What is the number of protons present in a unit length of the beam?

A $\frac{I}{e}$

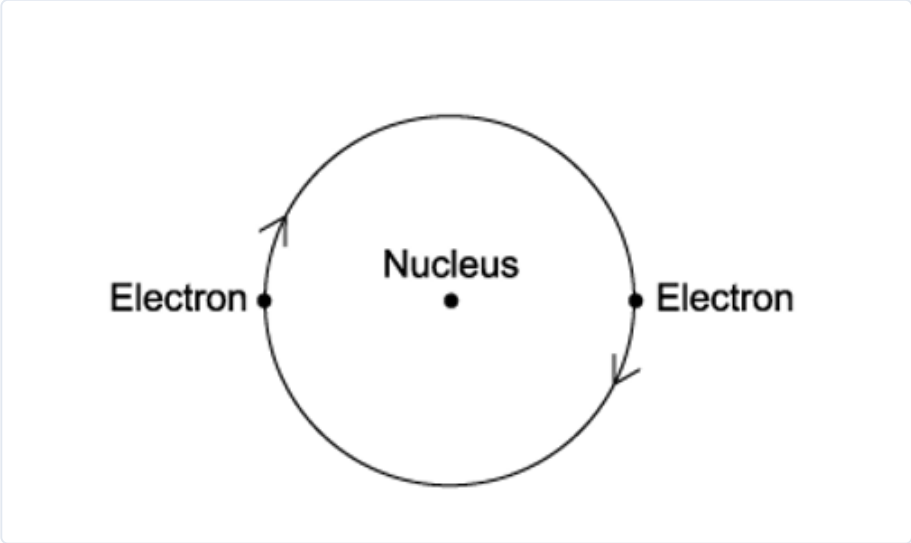
B $\frac{I}{ev}$

C $\frac{I}{e}v$

D $\frac{I}{v}$

Question 15 Hard [1 mark]

The diagram shows a model of an atom in which two electrons move round a circular orbit. The electrons complete 1.0×10^{15} revolutions around the nucleus every second.



What is the current caused by the motion of the electrons in the orbit?

A 1.6×10^{-34} A

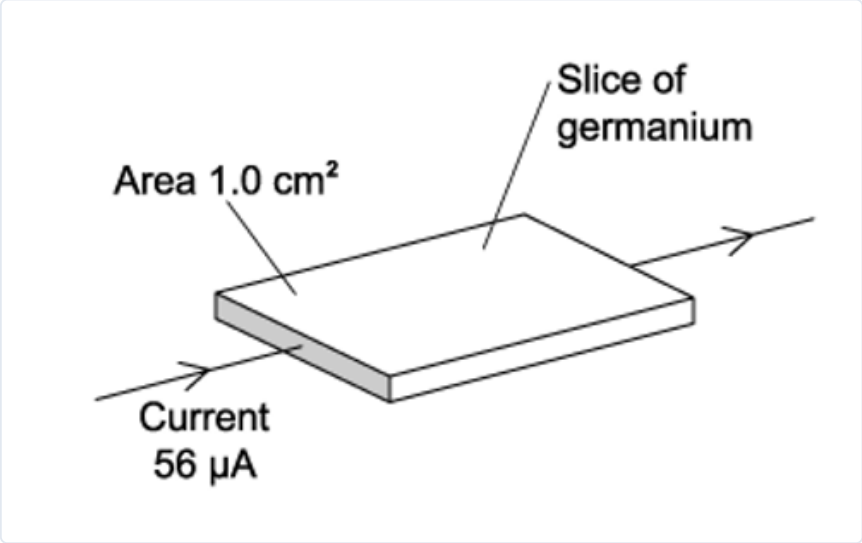
B 3.2×10^{-34} A

C 1.6×10^{-4} A

D 3.2×10^{-4} A

Question 16 Hard [1 mark]

A slice of germanium of cross-sectional area 1.0 cm^2 carries a current of 56 μA. The number density of charge carriers in the germanium is 2.0×10^{13} cm^{-3}.



Each charge carrier has a charge equal to the charge on an electron.

What is the average drift velocity of the charge carriers in the germanium?

A $0.18\ m\ s^{-1}$

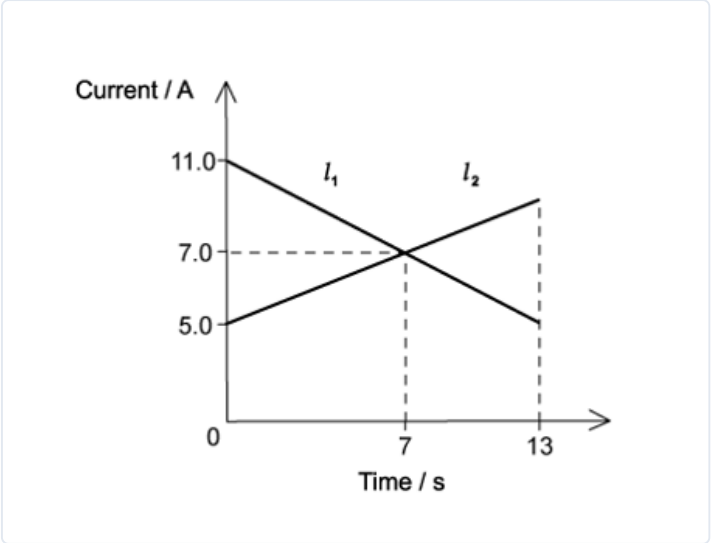
B $18\ m\ s^{-1}$

C $180\ m\ s^{-1}$

D $1800\ m\ s^{-1}$

Question 17 Hard [1 mark]

I_1 and I_2 are currents flowing through two different circuits, as shown in the diagram below.



Calculate the difference in the total charge that flows through the two circuits in the first 7 seconds.

A 0 C

B 21 C

C 42 C

D 49 C

Question 18 Hard [1 mark]

The Atlantic torpedo is a large electric fish capable of generating a voltage of 220 V between its tail and its head. This drives a pulse of current of 15 A lasting for a time of 2.0 ms. The fish produces 200 pulses per second.

What is the average power output of the fish?

A 33 W

B 1.3 kW

C 3.3 kW

D 6.6 kW

Question 19 Hard [1 mark]

A high potential is applied between the electrodes of an ionised gas. The gas carries a current of 8.16 mA and the number of electrons passing any point in the gas per unit time is $2.58 \times 10^{16}\ s^{-1}$

If the charge on each positive particle is 3.2×10^{-19} C, what is the number of positively charged particles passing any point in the gas per unit time?

A $1.26 \times 10^{16}\ s^{-1}$

B $2.58 \times 10^{16}\ s^{-1}$

C $3.84 \times 10^{16}\ s^{-1}$

D $10.3 \times 10^{16}\ s^{-1}$

Question 20 Hard [1 mark]

An electron beam of current 50 μA flows in a cathode-ray oscilloscope. The time-base causes the beam to sweep horizontally across the screen at 1.0×10^4 cm s^{-1}

What is the number of electrons arriving at the screen in one centimetre length of the horizontal trace?

A 3.1×10^{10}

B 3.1×10^{13}

C 1.3×10^{13}

D 8.0×10^{14}